

Week - 6:

WEIGHT INITIALIZATION - XAVIER
INITIALIZATION

FORWARD & BACKWARD PROPAGATION

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Machine Learning Lab

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Purpose of the lab:

To understand the Xavier (Glorot) Initialization, forward propagation, backward propagation and training the dataset generated.

To also be able to visualize the results.

Tasks performed:

Everything required has been performed.

Dataset Description:

Polynomial Type: QUADRATIC: $y = 0.98x^2 + 3.28x + 11.10$

Noise Level: $\varepsilon \sim N(0, 2.06)$

Architecture: Input(1) $\rightarrow$ Hidden(32) $\rightarrow$ Hidden(72) $\rightarrow$ Output(1)

Learning Rate: 0.005

Architecture Type: Narrow-to-Wide Architecture

All the result screenshots:

```
=====
PREDICTION RESULTS FOR x = 90.2
=====
```

```
Neural Network Prediction: 6,871.84
Ground Truth (formula):    8,253.53
Absolute Error:             1,381.70
Relative Error:             16.741%
```

```
=====
FINAL PERFORMANCE SUMMARY
=====
```

```
Final Training Loss: 0.074076
Final Test Loss:     0.074604
R2 Score:           0.9270
Total Epochs Run:    500
```

```
▶ print("Training Neural Network with your specific configuration...")
weights, train_losses, test_losses = train_neural_network(
    X_train_scaled, Y_train_scaled, X_test_scaled, Y_test_scaled,
    epochs=500, patience=10
)
```

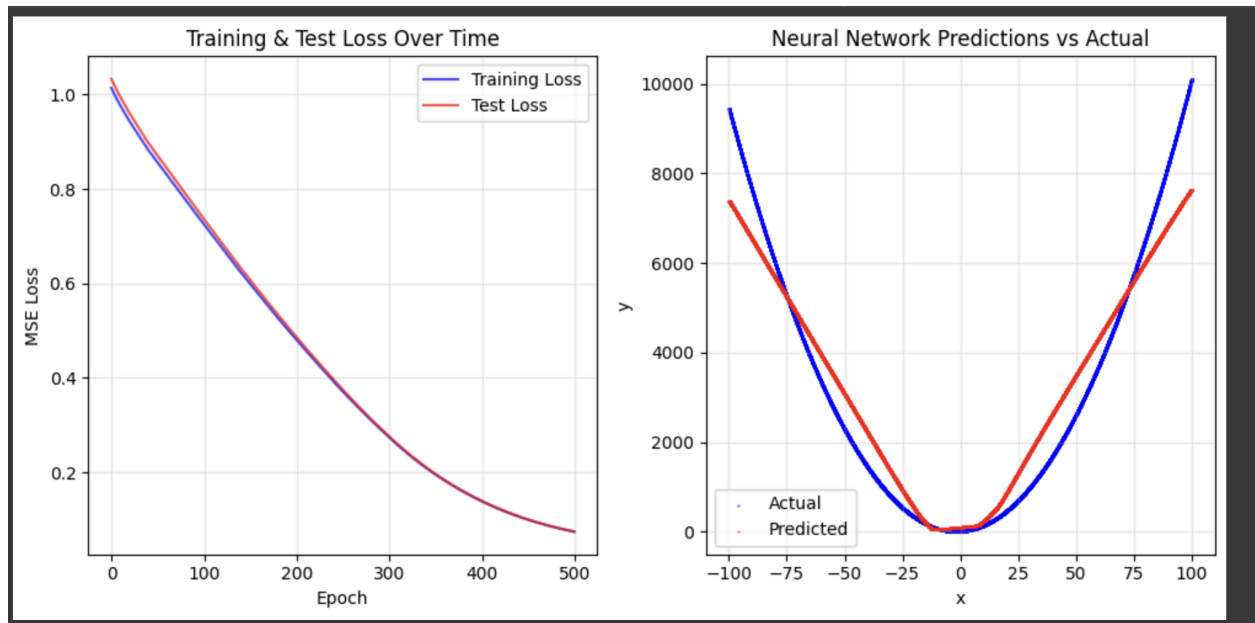
```
⇒ Training Neural Network with your specific configuration...
Starting training...
```

Architecture: 1 → 32 → 72 → 1

Learning Rate: 0.005

Max Epochs: 500, Early Stopping Patience: 10

```
-----
Epoch 20: Train Loss = 0.945234, Test Loss = 0.962363
Epoch 40: Train Loss = 0.884604, Test Loss = 0.900015
Epoch 60: Train Loss = 0.832116, Test Loss = 0.845919
Epoch 80: Train Loss = 0.779126, Test Loss = 0.791514
Epoch 100: Train Loss = 0.727283, Test Loss = 0.738475
Epoch 120: Train Loss = 0.675721, Test Loss = 0.685674
Epoch 140: Train Loss = 0.624792, Test Loss = 0.633766
Epoch 160: Train Loss = 0.576075, Test Loss = 0.583991
Epoch 180: Train Loss = 0.527971, Test Loss = 0.534850
Epoch 200: Train Loss = 0.481776, Test Loss = 0.487707
Epoch 220: Train Loss = 0.436944, Test Loss = 0.441970
Epoch 240: Train Loss = 0.393318, Test Loss = 0.397471
Epoch 260: Train Loss = 0.351533, Test Loss = 0.354961
Epoch 280: Train Loss = 0.312013, Test Loss = 0.314801
Epoch 300: Train Loss = 0.274729, Test Loss = 0.276930
Epoch 320: Train Loss = 0.240216, Test Loss = 0.241989
Epoch 340: Train Loss = 0.209585, Test Loss = 0.211032
Epoch 360: Train Loss = 0.182751, Test Loss = 0.183933
Epoch 380: Train Loss = 0.159255, Test Loss = 0.160231
Epoch 400: Train Loss = 0.138832, Test Loss = 0.139658
Epoch 420: Train Loss = 0.121226, Test Loss = 0.121945
Epoch 440: Train Loss = 0.106203, Test Loss = 0.106845
Epoch 460: Train Loss = 0.093513, Test Loss = 0.094101
Epoch 480: Train Loss = 0.082892, Test Loss = 0.083445
Epoch 500: Train Loss = 0.074076, Test Loss = 0.074604
```



Experiment	Xavier Initialization
Learning Rate	0.005
No of Epochs	500
Optimizer	Stochastic Gradient Descent
Activation Fn	ReLU
Final training loss	0.074076
Final test loss	0.074604
R^2 score	0.9270

Conclusion:

This gave my theoretical background of forward and backward propagation along with training datasets a strong practical understanding.

Thereby cementing my overall knowledge of this particular ML topic.